Hill Cipher

- Developed by the mathematician Lester Hill in 1929
- Strength is that it completely hides single-letter frequencies
  - The use of a larger matrix hides more frequency information
  - A 3 x 3 Hill cipher hides not only single-letter but also two-letter frequency information
- Strong against a ciphertext-only attack but easily broken with a known plaintext attack

Example of Hill Cipher

Let's walk through an example of encrypting the plaintext **"Hi"** using a **Hill Cipher** with a **2×2** matrix.

$$K = \begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix}$$

2. Convert plaintext to numbers

Use **A=0, B=1, ..., Z=25**.

H → 7

I → 8

So plaintext vector:

$$P = \begin{pmatrix} 7 \\ 8 \end{pmatrix}$$

Example of Hill Cipher

3. Encrypt: Multiply the matrix by the plaintext vector

Encryption formula:

$$C = (K \times P) \mod 26$$

Compute:

$$C = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \times \begin{bmatrix} 7 \\ 8 \end{bmatrix} = \begin{bmatrix} (3*7+3*8) \\ (2*7+5*8) \end{bmatrix} = \begin{bmatrix} (21+24) \\ (14+40) \end{bmatrix} = \begin{bmatrix} 45 \\ 54 \end{bmatrix}$$

Now take modulo 26:

$$C = \begin{bmatrix} 45 \mod 26 \\ 54 \mod 26 \end{bmatrix} = \begin{bmatrix} 19 \\ 2 \end{bmatrix}$$

4. Convert back to letters

- 19 → T
- 2 → C

Thus, the ciphertext is "**TC**".

Example of Hill cipher

- This example will rely on some linear algebra and some number theory. The *key* for a hill cipher is a matrix e.g.

$$\begin{bmatrix} 2 & 4 & 5 \\ 9 & 2 & 1 \\ 3 & 17 & 7 \end{bmatrix}$$

- In the above case, we have taken the size to be 3×3, however it can be any size (as long as it is square).

- Assume we want to encipher the message **ATTACK AT DAWN**. To encipher this, we need to break the message into chunks of 3. We now take the first 3 characters from our plaintext, **ATT** and create a vector that corresponds to the letters (replace A with 0, B with 1 ... Z with 25 etc.) to get: [0 19 19] (this is ['A' 'T' 'T']).

Example of Hill Cipher

Step 1: Prepare the Message

Remove spaces: "ATTACKATDAWN"

Convert letters to numbers (A=0, B=1, ..., Z=25):

A T T A C K A T D A W N

0 19 19 0 2 10 0 19 3 0 22 13

Step 2: Break into 3-letter Groups

We have four groups:

[0, 19, 19] (ATT)

[0, 2, 10] (ACK)

[0, 19, 3] (ATD)

[0, 22, 13] (AWN)

Note: If the last group isn't complete, we'd pad it with X (23), but ours is complete.

Example of Hill Cipher

Step 3: Encrypt Each Group Encryption

formula: C = K × P mod 26

Group 1: [0, 19, 19]

[2 4 5] [0] [2*0 + 4*19 + 5*19] [0 + 76 + 95] [171] [171 mod 26] [15] (P)

[9 2 1] × [19] = [9*0 + 2*19 + 1*19] = [0 + 38 + 19] = [57] = [57 mod 26] = [5](F)

[3 17 7] [19] [3*0 +17*19 +7*19] [0 +323 +133] [456] [456 mod 26] [14(O)

First ciphertext group: [15, 5, 14] → "PFO"

Group 2: [0, 2, 10]

[2 4 5] [0] [2*0 + 4*2 + 5*10] [0 + 8 + 50] [58] [58 mod 26] [6] (G)

[9 2 1] × [2] = [9*0 + 2*2 + 1*10] = [0 + 4 + 10] = [14] = [14 mod 26] = [14] (O)

[3 17 7] [10] [3*0 +17*2 +7*10] [0 +34 +70] [104] [104 mod 26] [0] (A)

Second ciphertext group: [6, 14, 0] → "GOA"

Example of Hill Cipher

Group 3: [0, 19, 3]

[2 4 5] [0] [2*0 + 4*19 + 5*3] [0 + 76 + 15] [91] [91 mod 26] [13] (N)
[9 2 1] × [19] = [9*0 + 2*19 + 1*3] = [0 + 38 + 3] = [41] = [41 mod 26] = [15] (P)
[3 17 7] [3] [3*0 +17*19 +7*3] [0 +323 +21] [344] [344 mod 26] [6] (G)

Third ciphertext group: [13, 15, 6] → "NPG"

Group4: [0,22,13]

[2 4 5] [0] [2*0 + 4*22 + 5*13] [0 + 88 + 65] [153] [153 mod 26] [23] (X)
[9 2 1] × [22] = [9*0 + 2*22 + 1*13] = [0 + 44 + 13] = [57] = [57 mod 26] = [5] (F)
[3 17 7] [13] [3*0 +17*22 +7*13] [0 +374 +91] [465] [465 mod 26] [23] (X)

Fourth ciphertext group: [23, 5, 23] → "XFX“

Final Ciphertext

Combining all ciphertext groups: "PFO GOA NPG XFX“

Encrypted Message:- “PFPGOANPGXFX”

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